

1. What are the key differences between SQL and NoSQL databases?

SQL	NoSQL
Relational database	Non-relational database
Uses tables, rows, columns	Uses documents, key-value, graph, or column-family models
Fixed schema	Flexible schema
Uses SQL language	Uses various query languages
Vertical scaling	Horizontal scaling
Best for structured data	Best for semi-structured and unstructured data

2. What makes MongoDB a good choice for modern applications?

- Flexible document-based schema.
 - High scalability through sharding.
 - High availability through replication.
 - Fast development and deployment.
 - Handles large volumes of structured and unstructured data.
 - Supports cloud deployment through MongoDB Atlas.
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3. Explain the concept of collections in MongoDB.

A **collection** is a group of MongoDB documents, similar to a table in a relational database. Collections do not require a fixed schema, allowing documents with different structures to exist within the same collection.

4. How does MongoDB ensure high availability using replication?

MongoDB uses **Replica Sets**, where:

- One node acts as the **Primary**.
 - Other nodes act as **Secondary** replicas.
 - Data is copied from the primary to secondary nodes.
 - If the primary fails, a secondary node is automatically elected as the new primary.
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5. What are the main benefits of MongoDB Atlas?

- Fully managed cloud database service.
 - Automatic backups.
 - Built-in security features.
 - Auto-scaling capabilities.
 - Global cluster deployment.
 - Performance monitoring and analytics.
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6. What is the role of indexes in MongoDB, and how do they improve performance?

Indexes improve query performance by allowing MongoDB to locate data quickly without scanning the entire collection. They reduce query execution time and improve sorting and filtering operations.

7. Describe the stages of the MongoDB aggregation pipeline.

Common aggregation stages include:

- **\$match** – Filters documents.
 - **\$group** – Groups documents and performs calculations.
 - **\$project** – Selects or modifies fields.
 - **\$sort** – Sorts documents.
 - **\$limit** – Limits the number of results.
 - **\$lookup** – Joins data from another collection.
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8. What is sharding in MongoDB? How does it differ from replication?

Sharding distributes data across multiple servers to improve scalability.

Replication copies data across multiple servers to improve availability and fault tolerance.

Sharding	Replication
Improves scalability	Improves availability
Splits data	Copies data
Distributes workload	Provides backup/failover

9. What is PyMongo, and why is it used?

PyMongo is the official Python driver for MongoDB. It allows Python applications to connect, query, insert, update, and delete data in MongoDB databases.

10. What are the ACID properties in the context of MongoDB transactions?

ACID stands for:

- **Atomicity** – All operations succeed or none do.
 - **Consistency** – Data remains valid before and after transactions.
 - **Isolation** – Concurrent transactions do not interfere with each other.
 - **Durability** – Committed changes are permanently stored.
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11. What is the purpose of MongoDB's explain() function?

The **explain()** function provides information about how MongoDB executes a query, including:

- Query execution plan.
 - Index usage.
 - Performance statistics.
 - Query optimization details.
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12. How does MongoDB handle schema validation?

MongoDB uses **JSON Schema Validation** rules to enforce data structure requirements. Validation rules can specify required fields, data types, value ranges, and other constraints.

13. What is the difference between a primary and a secondary node in a replica set?

Primary Node	Secondary Node
Accepts write operations	Usually handles read operations
Source of replication	Replicates data from primary
Only one primary exists	Multiple secondary nodes can exist

14. What security mechanisms does MongoDB provide for data protection?

MongoDB provides:

- Authentication.
 - Role-Based Access Control (RBAC).
 - TLS/SSL encryption.
 - Encryption at rest.
 - Auditing.
 - IP whitelisting.
 - Data masking and access controls.
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15. Explain the concept of embedded documents and when they should be used.

Embedded documents store related data within a single document.

Example:

```
{
  "name": "John",
  "address": {
    "city": "Delhi",
    "state": "Odisha"
  }
}
```

Use embedded documents when:

- Data is frequently accessed together.
 - One-to-one or one-to-few relationships exist.
 - Faster reads are required.
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16. What is the purpose of MongoDB's \$lookup stage in aggregation?

\$lookup performs a left outer join between collections, allowing related data from another collection to be combined into the aggregation result.

17. What are some common use cases for MongoDB?

- E-commerce applications.
 - Content management systems.
 - Real-time analytics.
 - Social networking platforms.
 - IoT applications.
 - Mobile and web applications.
 - Big data applications.
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18. What are the advantages of using MongoDB for horizontal scaling?

- Handles large datasets efficiently.
 - Supports sharding.
 - Distributes workload across servers.
 - Improves performance and availability.
 - Allows seamless growth without major architecture changes.
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19. How do MongoDB transactions differ from SQL transactions?

- SQL databases traditionally support multi-table ACID transactions.
 - MongoDB originally focused on document-level operations but now supports multi-document ACID transactions.
 - MongoDB transactions are designed for distributed and document-oriented environments.
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20. What are the main differences between capped collections and regular collections?

Capped Collections	Regular Collections
Fixed size	Dynamic size
Maintain insertion order	No guaranteed order
Old documents overwritten when full	Documents remain until deleted
Faster for logging data	General-purpose storage

21. What is the purpose of the \$match stage in MongoDB's aggregation pipeline?

The **\$match** stage filters documents based on specified conditions. It works similarly to the `find()` method and reduces the number of documents processed in later stages.

22. How can you secure access to a MongoDB database?

- Enable authentication.
 - Use strong passwords.
 - Implement RBAC.
 - Enable TLS/SSL encryption.
 - Restrict network access with firewalls.
 - Use IP whitelisting.
 - Regularly update MongoDB versions.
 - Monitor and audit database activities.
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23. What is MongoDB's WiredTiger storage engine, and why is it important?

WiredTiger is MongoDB's default storage engine. It provides:

- High-performance reads and writes.
- Compression for efficient storage.
- Document-level concurrency control.
- Improved memory management.
- Better scalability and reliability.